ANALYTICAL CUMULATIVE EXAMINATION

April 6, 1991

1. (50 points) Bi^{+3} forms a complex with xylenol orange (XO). Various molar ratios of Bi^{+3} to XO were prepared in 0.1 N H_2SO_4 and the absorbance of the solutions measured at 545 nm, a wavelength where only the complex absorbs. The data from these experiments is listed below.

a. Plot the data and determine the molar ratio of Bi^{+3} to XO in the complex.

b. Estimate the formation constant of the Bi^{+3}- XO complex noting that $[Bi^{+3}] + [XO] = 2.4 \times 10^{-5}$ M.

$[Bi^{+3}] / [Bi^{+3}] + [XO]$	Corrected Absorbance
0.0	0.00
0.1	0.05
0.2	0.10
0.3	0.15
0.4	0.18
0.5	0.20
0.6	0.19
0.7	0.16
0.8	0.11
0.9	0.06
1.0	0.00

2. (50 points) Molar absorptivities of absorbing molecules can be used to obtain compositional information for derivatives of the parent molecule (e.g. molecular weight) providing the added group does not affect the chromophore. For example, literature values for the molar absorptivities of nucleotides are often used to determine the concentration of nucleotide solutions, since even highly purified nucleotides may contain variable waters of hydration or a variable salt content. A sample of the disodium salt of cytidine 5'-monophosphate (5'-CMP, $C_9H_{12}N_3O_8PNa_2$) was weighed out (0.0814 g), dissolved and diluted to 25.00 mL in a volumetric flask with distilled water. ε for 5'-CMP is 1.30×10^4 at λ_{max} = 280 nm in 0.01 M HCl. 0.10 mL of the CMP stock solution was diluted to 10.00 mL with 0.01 M HCl. The absorbance of this solution was measured as 0.838 in a 1.0 cm cuvette.

a. Calculate the expected concentration of 5'-CMP assuming it to be a pure, anhydrous salt.

b. Calculate the concentration based on the spectrophotometric measurement.

c. Assuming the difference in concentration is due only to absorbed and bound water molecules, calculate the average number of water molecules per 5'-CMP molecule.